

Lista de problemas para el primer parcial

Ex1 (Spanning trees)

Say and prove whether the following problems are in P or are NP-complete. Given a graph $G = (V, E)$ together with a set of vertices $S \subseteq V$,

- (a) Find a spanning tree such that its leaf set includes S .
- (b) Find a spanning tree such that the set of leaves is exactly S
- (c) Find a spanning tree such that the set of leaves is a subset of S .

A *spanning tree* is a subgraph of G that (1) it is a tree and (2) its vertex set is V .

A *leaf* is a vertex with degree 1.

You can use the fact that the Hamiltonian path problem, given a graph and two vertices a and b , has G a Hamiltonian path starting at a and ending at b ?, is NP-complete.

A path is Hamiltonian if it is a simple path (no vertex is repeated) and it passes through all the vertices in the graph.

Ex2 (Color)

Consider the problem COLOR:

INPUT: Undirected graph $G = (V, E)$ and $k \in \mathbb{N}$.

Question: Is there an assignment $c : V \rightarrow \{1, \dots, k\}$ such that, for each $(u, v) \in E$, $c(u) \neq c(v)$.

- (a) Show that COLOR belongs to NP.

Consider the following construction that transforms an instance of SAT into a graph:

Let ϕ a boolean formula in CNF on variables $x_1, x_2, \dots, x_n$ and clauses $C_1, C_2, \dots, C_m$. From ϕ we construct the graph $f(\phi) = (V, E)$ as follows,

$$\begin{aligned} V &= \{v_1, \dots, v_n\} \cup \{x_1, \dots, x_n\} \cup \{\bar{x}_1, \dots, \bar{x}_n\} \cup \{c_1, \dots, c_m\} \\ E &= \{\{v_i, v_j\} : \forall i, j, 1 \leq i, j \leq n, \text{ amb } i \neq j\} \\ &\quad \cup \{\{v_i, x_j\}, \{v_i, \bar{x}_j\} : \forall i, j, 1 \leq i, j \leq n \text{ amb } i \neq j\} \\ &\quad \cup \{\{x_i, \bar{x}_i\} : \forall i, 1 \leq i \leq n\} \\ &\quad \cup \{\{x_i, c_l\} : \forall i, 1 \leq i \leq n, \text{ on } x_i \text{ es un literal a } C_l\} \\ &\quad \cup \{\{\bar{x}_i, c_l\} : \forall i, 1 \leq i \leq n, \text{ on } \bar{x}_i \text{ no es un literal a } C_l\} \end{aligned}$$

- (b) Draw the graph resulting after applying f to

$$(\bar{x}_1 \vee x_2) \wedge (x_1 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee \bar{x}_3)$$

- (c) With the help of f , show that $\text{SAT} \leq_m^p k\text{-Color}$.

Ex3 Geography game.

One way to pass the time during long journey is to play the word game of *Geography*. The first player names a place, say Barcelona. The second player replies with a place whose name starts with the same letter the previous place ended with, such as Athens. The first player could then respond with Sicily, and so on. Each place can only be named once, and the first player who cannot make a move loses.

We can make a mathematical version of this game. We have a directed graph $G = (V, E)$, and a designated start node $s \in V$. The nodes represent the places and the directed edges represent the legal moves. We start at s , and you and I take turns deciding which step to do next. Whoever ends up in a dead end, where every outgoing edge points to a vertex we have already visited, loses. A position or node v is a *winning position* for the first player if whatever legal move the second player selects, the first player never ends up in a dead end.

GEOGRAPHY

Input: An directed graph $G = (V, E)$ and a start node s .

Question: Is s a *forced* win position for the first player? (The first player with strategy s always wins whatever is the strategy selected by the second player).

Show that GEOGRAPHY is in PSPACE (In fact GEOGRAPHY is a well known PSPACEcomplete problem).

Ex4 Independent set.

Consider a group of n people. Suppose that the total number of acquaintances (pairs of people which know each other) in the group is $2n$. We model this situation as an undirected graph $G = (V, E)$, in which the vertices are people and we draw an edge between two people if and only if they know each other. The total number of edges is $2n$. In this problem, our aim is to find a fairly large group of people all of whom are strangers. Such a group of people is called an *independent set*.

- (a) We construct a random subset $S \subseteq V$ of people as follows: for each person we independently flip a bias coin with $\Pr[H] = 1/c$, for some constant $c > 1$. If the flip comes up heads, we put the person in S .

What is the expected number of people in S ? (i.e., $\mathbf{E}[|S|]$)

- (b) Let X be a r.v. which counts the number of edges inside S (i.e., edges of the graph that connect pairs of people in S).

What is the value of $\mathbf{E}[X]$?

- (c) Now for each $e \in S$, choose arbitrarily an endpoint of e and remove it.

Find an upper bound to $\mathbf{E}[|S'|]$.

- (d) Argue whether S' is or is not an independent set.

- (e) Using the previous items, give an efficient randomized algorithm that, with probability at least $1/15$, finds an independent set of at least $n/16$ people.

Ex6 (Pattern matching)

Given a text $T = x_1x_2 \cdots x_n$ and a pattern $S = s_1 \cdots s_m$, with $m \ll n$ and both chains over the same alphabet $\Sigma = \{0, 1\}$, we want to determine if S occurs as a contiguous substring of T . For ex., if $T = 10110110010101110$ and $S = 1011$ then $101\boxed{1011}0010\boxed{1011}10$. The standard greedy takes $O(nm)$ steps. There are deterministic algorithms that work in worst-time $O(n + m)$ (Knuth-Morris-Pratt), but they are complicated, difficult to implement and with large implementation constants. The following simple probabilistic algorithm does the job using the fingerprint technique, with a small probability of error. The algorithm computes the fingerprint of S and compares with the fingerprints of successive sliding substring of T , i.e. with $T(j) = x_j \cdots x_{j+m-1}$, for $1 \leq j \leq n - m + 1$.

Matching P, T

Express S as an integer $D(S) = \sum_{i=0}^{m-1} x_{i+1}2^i$

so $D(S)$ is a m -bit integer

Choose a prime $p \in [2, \dots, k]$,

where $k = cmn \ln(cmn)$, for suitable $c > 1$

Compute $\phi(S) = D(S) \bmod p$

for $j = 1$ to $n - m + 1$ **do**

 Compute $D(T(j)) = \sum_{i=0}^{m-1} x_{j+i}2^i$

 Compute $\phi(T(j)) = D(T(j)) \bmod p$

if $\phi(T(j)) = \phi(S)$ **then**

output match at position j

endif

endfor

Prove,

- (a) This algorithm is one-side, it may output match when there is no match. Prove that $\Pr[\text{output match, when no match}] \leq 1/c$, for suitable $c > 0$.
- (b) Prove that the algorithm can be implemented in $O(n + m)$ steps.

Ex6 (Words)

Consider the following problem whose input is a $n \times n$ grid, where each *square* contains a letter. Also, given as a part of the input, there is a database with $m < n^2$ words, all the words have the same length ℓ , so the input's size is $(n^2 + m\ell)$. Each one of the words in the database has been placed in an horizontal or vertical position in the grid (in ℓ consecutive squares, as in the database or in reversed order). In this version of the problem words can't be in diagonal and the intersection of two words is empty. The remaining positions of the square have been filled arbitrary with letters.

The example below has $n = 8, w = 7, l = 4$ with data base (VUIT, HASH, ROCA, TRES, TRET, CLAU, KILO).

L	A	S	B	A	R	C	A
A	A	F	D	H	S	A	H
R	O	C	A	Z	U	C	E
I	T	T	I	U	V	U	K
S	I	L	A	L	B	A	I
A	C	O	B	U	K	L	L
L	B	T	E	R	T	C	O
B	E	L	T	R	E	S	Z

Design an expected linear time randomized algorithm to find the positions in the grid of all the mots from the database. For each word, the algorithm must return the coordinates of the square, where it starts. Justify the time complexity of your algorithm as a function of n, m, ℓ .